

Olivier Belan

PhD Student

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TECHNICAL SKILLS

Languages: Python, Cython, C++, C, C#, Java

Libraries/Frameworks: Pytorch, Numba, Jax, Numpy, Cupy, Cuda C++, Ray

Developer Tools: Git, Docker

OPEN PROJECT

Evo-Sim

github.com/OlivierBelan/Evo-Sim

- Evo-Sim, developed from scratch, simulator for Spiking and Artificial Neural Networks, optimized for CPU multi-threading and GPU. Supports diverse evolutionary simulation algorithms and neural network techniques, including GA, NEAT, HyperNEAT, MAP-ELITE, NSLC, CMA-ES, NES, OpenES, ARS, or ES-HyperNEAT.

SNN-Sim

github.com/OlivierBelan/SNN-Sim

- SNN-Sim, an advanced Spiking Neural Network simulator designed in Cuda, Cython and Python, specifically optimized for evolutionary techniques and GPU/CPU multi-threading. It supports various learning paradigms, including supervised and reinforcement learning, and is compatible/optimized with NeuroEvolution algorithms such as NEAT, CMA-ES, NES, among others.

EXPERIENCE

Research Assistant

International Centre for Neuromorphic Systems (ICNS)

March. 2025 –

Sydney, Australia

- Research Assistant at ICNS under the supervision of Dr Saeed Afshar, Dr Yeshwanth Ravi Theja Bethi, Dr Alexandre Marcireau and Dr Sergio Chevtchenko, Focuses on Spiking Neural Networks, NeuroEvolution, and Agent Control. Development of SNNs tailored for real-time edge applications

Research Assistant & Software Developer

University of Kent

Jan. 2025 –

Canterbury, United Kingdom

- Personalising Domiciliary Dementia Care: ML Researcher and Developer for App Logic and Personalised Care Matching, under the supervision of Dr Daniel Soria

Researcher & Graduate Teaching Assistant

University of Kent

Sep. 2021 –

Canterbury, United Kingdom

- Artificial Intelligence and Data Analytics (AIDA) Group: As a contributing member, I engage in research and projects aimed at pushing the boundaries of AI and data analytics, focusing on developing cutting-edge solutions.
- Teaching Assistant: Leading classes for both undergraduate and postgraduate students, covering essential modules such as Cognitive Neural Networks, Machine Learning, DeepLearning, Programming for AI, Advanced OOP and Functional Programming.

Research and Development - Software Developer

Pasteur Institute and Brain & Spine Institute

Jun 2020 - Feb 2022

Paris, France

- Toi-même: An ecosystem for mental disorder research, in collaboration with the Pasteur Institute and Brain and Spine Institute, Paris. My role involved to enhance the software tools of the Toi-même Solution.

Teaching Assistant

Web@cademie by Epitech

Sep 2019 - Mar 2020

Paris, France

- Teaching in Programming and Project Management, developing students professionalism and leadership abilities. Actively supported first and second-year undergraduates towards achieving their career aspirations.

Teaching Assistant

Sep 2019 - Mar 2020

Coding academy by EPITECH

Paris, France

- Role reflected responsibilities similar to those at Web@cademie, albeit for a different institution with different student profiles.

Software Developer

Jul 2018 - Dec 2018

Dt Consulting

Paris, France

- Tasked with contributing to the product roadmap for a C# ticketing solution tailored to sports venues, enhancing functionality and user experience.

PAPERS

Fragmented Fourier Series to optimise Spiking Neural Network

University of Kent

Canterbury, United Kingdom

Neuron Energy Distribution to optimise Spiking Neural Network

University of Kent

Canterbury, United Kingdom

Spikes Augmentation Decoder for Spiking Neural Network

University of Kent

Canterbury, United Kingdom

Combinatorial Encoder for Spiking Neural Network

University of Kent

Canterbury, United Kingdom

Benchmark: NeuroEvolution and Spiking Neural Network in RL

University of Kent

Canterbury, United Kingdom

EDUCATION

University of Kent

Canterbury, United Kingdom

Philosophiae Doctor, School of Computing

2021 -

- Conducting research on NeuroEvolution and Spiking Neural Networks within Neuromorphic systems for Reinforcement Learning challenges, guided by Dr. Dominique Chu and Dr. Marek Grzes.

University of Kent

Canterbury, United Kingdom

Master of Science in Advanced Computer Science (Computational Intelligence)

2020 - 2021

- Master thesis: NeuroEvolution and Spiking Neural Network (Distinction), Supervised by Dr. Dominique Chu
- MSc specialized curriculum focusing on Artificial Intelligence (Distinction)

Epitech

Paris, France

School of Computing

2020 - 2021

- Final Project and Dissertation: "Polaris/Toi-même" - Developed an ecosystem for mental disorder research, in collaboration with the Pasteur Institute and Brain and Spine Institute, Paris. Achieved finalist status in a nationwide competition among Epitech schools - (Distinction).

Epitech

Paris, France

Bachelor's Degree, School of Computing

2017 - 2020

- Specialized in low-level programming, with a focus on developing a strong foundational skill set in C and C++ languages.

CONFERENCES

NCN, Neuromorphic Computing Netherlands (poster)

September 2024

Eindhoven University of Technology

IEEE/ACM ICONS, International Conference On Neuromorphic Systems (poster)

August 2024

George Mason University

FUNDING & SCHOLARSHIP

PhD Scholarship - Graduate Teaching Assistant - 20 000£

University of Kent

Sep 2021 - Sep 2024

Canterbury, United Kingdom

SERVICE AND VOLUNTARISM

UKC French Society - Head of Logistic

University of Kent

2023 - 2024

Canterbury, United Kingdom

- The French Society at the University of Kent serves as a vibrant hub for those interested in French culture, whether they're fluent speakers or just beginning to delve into the language.

Our aim was to promote and celebrate the richness of French traditions through a diverse array of events. From culinary showcases of French cuisine to French cinema nights and insightful discussions on literature, we offer a myriad of opportunities for members to connect, learn, and immerse themselves.

As Head of Logistics, I planned and executed events, managed resources, liaised with vendors, and ensured memorable experiences for attendees

Coding Club By Epitech - Teaching

Epitech

2017 - 2020

Paris, France

- The Coding Club is a recurrent workshop that introduces programming and development. The sessions are led in a fun way through various activities, such as developing video games, solving puzzles or cyber security activities, and more!

LANGUAGES & PASSION

Languages: French (native), English (Fluent)

Passion: Apéro, Wine, Martial Arts, Rugby, Football (Soccer)

REFERENCES

Dominique Chu: d.f.chu@kent.ac.uk

Howard Bowman: bowmanh@adf.bham.ac.uk

Fabrice Marco: fabrice.marco@epitech.eu

Marek Grzes: m.grzes@kent.ac.uk

Daniel Soria: d.soria@kent.ac.uk

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Alexandre Marcireau : a.marcireau@westernsydney.edu.au

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